

OCTOBER 2026

Predictive Mock Test SOLUTION BOOK

MATHS 2 + BIOLOGY PATHWAY

81

questions

3

modules

40

minutes per module

MODULES IN THIS BOOK

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MODULE 01

Mathematics 1

QUESTIONS & WORKED SOLUTIONS

Maths 2 + Biology pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

Using this module

Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

A rectangular card has a length recorded as 12 cm and a width recorded as 8 cm. Both measurements are rounded to the nearest centimetre. What is the least upper bound for the ratio of its actual length to its actual width?

A $\frac{3}{2}$

B $\frac{5}{3}$

C $\frac{23}{15}$

D $\frac{25}{17}$

E $\frac{25}{16}$

Worked solution

Answer B

Write the actual length and width as L and W . Their bounds are $11.5 \leq L < 12.5$ and $7.5 \leq W < 8.5$. The largest ratios use a length approaching 12.5 and the smallest width, 7.5. Thus $\frac{L}{W} < \frac{12.5}{7.5} = \frac{5}{3}$, and ratios can approach $\frac{5}{3}$ as closely as desired. The least upper bound is $\frac{5}{3}$.

Trap: using both upper bounds. To make a positive ratio as large as possible, make its numerator large and its denominator small. The upper bound need not be attained.

Question 02

For a real number k , the equation $x^2 - (k + 4)x + 2k = 0$ has one root strictly between 0 and 1, and its other root is greater than 2. What is the set of all possible values of k ?

A $k < 0$

B $0 \leq k \leq 3$

C $k > 3$

D $0 < k < 3$

E $k < 0$ or $k > 3$

Worked solution

Answer D

Let $p(x) = x^2 - (k + 4)x + 2k$. A root in $(0, 1)$ requires $p(0) > 0$ and $p(1) < 0$, giving $k > 0$ and $k < 3$. Also $p(2) = -4 < 0$, so after the first crossing the other root is beyond 2.

Trap: checking only the discriminant and forgetting where the roots lie.

Question 03

What is the value of x that satisfies $\frac{9^{x+1} 27^{2-x}}{3^{x-1}} = 81$?

A $\frac{5}{2}$

B $\frac{3}{2}$

C 2

D $\frac{1}{2}$

E $\frac{7}{2}$

Worked solution

Answer A

Write every number as a power of 3. The equation becomes $3^{2(x+1)+3(2-x)-(x-1)} = 3^4$.
Hence $9 - 2x = 4$, giving $x = \frac{5}{2}$.

Trap: subtract the whole denominator exponent $x - 1$, so its contribution is $-x + 1$.

Question 04

The least positive integer n leaves remainder 2 when divided by 5, and remainder 3 when divided by 7. What is the remainder when $n^2 + 4n + 1$ is divided by 35?

A 3

B 5

C 18

D 13

E 8

Worked solution

Answer E

The candidates leaving remainder 2 on division by 5 are 2, 7, 12, 17, The first that leaves remainder 3 on division by 7 is $n = 17$. Then $n^2 + 4n + 1 = 358 = 10 \cdot 35 + 8$, so the remainder is 8.

Question 05

A container holds 4 red, 3 blue and 2 green counters. Three counters are drawn without replacement. Given that exactly one of the three is green, what is the probability that the first counter drawn is red?

A $\frac{1}{7}$

B $\frac{8}{21}$

C $\frac{2}{9}$

D $\frac{4}{9}$

E $\frac{1}{2}$

Worked solution

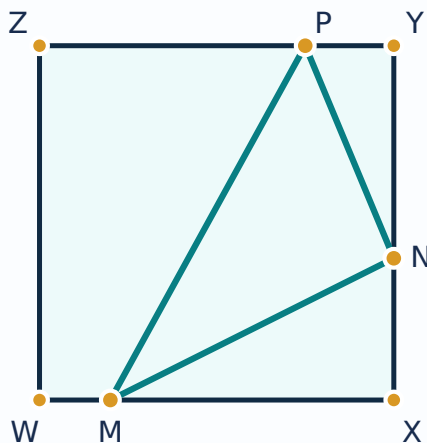
Answer B

Given that exactly one of the three counters is green, the green counter is equally likely to occupy any of the three positions. Therefore the first counter is non-green with probability $\frac{2}{3}$. Conditional on its being non-green, each of the seven non-green counters is equally likely to be first, and four are red. Hence the required probability is $\frac{2}{3} \times \frac{4}{7} = \frac{8}{21}$.

Trap: conditioning changes the sample space; the unconditional probability $\frac{4}{9}$ is not the answer.

Question 06

$WXYZ$ is a square of side 1. Points M, N, P lie on WX, XY, YZ , respectively, with $WM : MX = 1 : 4$, $XN : NY = 2 : 3$, and $YP : PZ = 1 : 3$. What is the area of triangle MNP ?



Schematic diagram; not to scale.

A $\frac{1}{5}$

B $\frac{1}{4}$

C $\frac{29}{100}$

D $\frac{3}{10}$

E $\frac{7}{20}$

Worked solution

Answer C

The ratios give $WM = \frac{1}{5}$, $MX = \frac{4}{5}$, $XN = \frac{2}{5}$, $NY = \frac{3}{5}$, $YP = \frac{1}{4}$, and $PZ = \frac{3}{4}$. Subtract the two right triangles MXN , NYP , and the trapezium $WMPZ$ from the unit square.

The two triangle areas are $\frac{1}{2} \cdot \frac{4}{5} \cdot \frac{2}{5} = \frac{4}{25}$ and $\frac{1}{2} \cdot \frac{3}{5} \cdot \frac{1}{4} = \frac{3}{40}$. The trapezium has parallel sides WM and ZP , separated by distance 1, so its area is $\frac{1}{2}(\frac{1}{5} + \frac{3}{4}) = \frac{19}{40}$. Thus the required area is $1 - \frac{4}{25} - \frac{3}{40} - \frac{19}{40} = \frac{29}{100}$.

Tip: subtract familiar shapes from the square; no general coordinate-area formula is needed.

Question 07

Pumps A and B fill a tank at constant rates. In a trial, increasing A's rate by 20% and decreasing B's rate by 10% leaves their combined rate unchanged.

At its original rate, A alone fills the tank in 30 minutes. The empty tank is now filled using both pumps at their original rates. After 4 minutes, B is switched off and A completes the filling.

What is the total time taken to fill the tank?

A 14 minutes

B 18 minutes

C 20 minutes

D 22 minutes

E 24 minutes

Worked solution

Answer D

Let the original filling rates be a and b tanks per minute. The trial gives

$$1.2a + 0.9b = a + b \Rightarrow 0.2a = 0.1b \Rightarrow b = 2a.$$

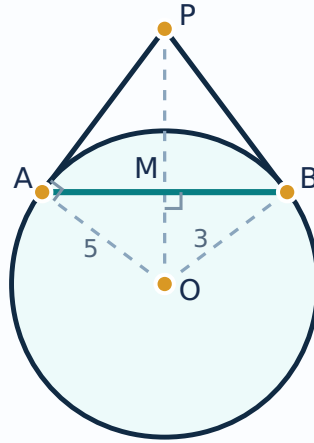
Since $a = 1/30$, the combined rate is $3a = 1/10$ tank per minute. In the first 4 minutes, the pumps fill $4/10 = 2/5$ of the tank.

A fills the remaining $3/5$ in $(3/5) \times 30 = 18$ minutes. The total is $4 + 18 = 22$ minutes.

Useful approach: Use the unchanged combined rate to find the ratio of the two original rates before calculating any times.

Question 08

A circle has centre O and radius 5. Chord AB is 3 units from O , measured perpendicularly. The tangents at A and B meet at P . What is OP ?



Schematic diagram; M is the midpoint of AB .

A $\frac{25}{3}$

B $\frac{25}{6}$

C 5

D 10

E 15

Worked solution

Answer A

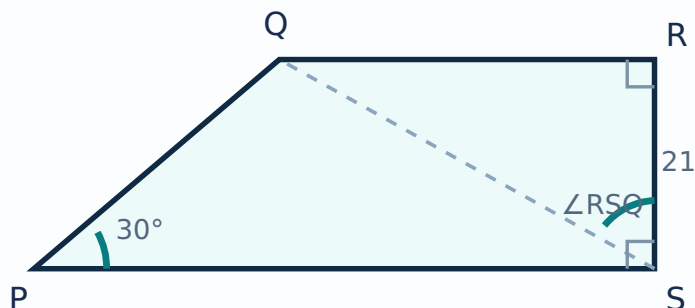
Let M be the midpoint of chord AB . The perpendicular from O to AB passes through M , and O, M, P are collinear by symmetry. We have $OM = 3$ and $OA = 5$.

Triangle OMA is right-angled at M , while triangle OAP is right-angled at A , because a radius is perpendicular to a tangent. They share the angle at O , so they are similar.

Hence $\frac{OM}{OA} = \frac{OA}{OP}$, giving $\frac{3}{5} = \frac{5}{OP}$ and $OP = \frac{25}{3}$.

Question 09

$PQRS$ is a right trapezium with $PS \parallel QR$, right angles at R, S , $RS = 21$, $\angle QPS = 30^\circ$, and $\tan \angle RSQ = \frac{4}{3}$. What is PS ?



Schematic diagram; diagonal SQ is shown for the given angle.

A $28 + 21\sqrt{3}$

B $21 + 21\sqrt{3}$

C $21 + 28\sqrt{3}$

D $49 + \sqrt{3}$

E $28 + 7\sqrt{3}$

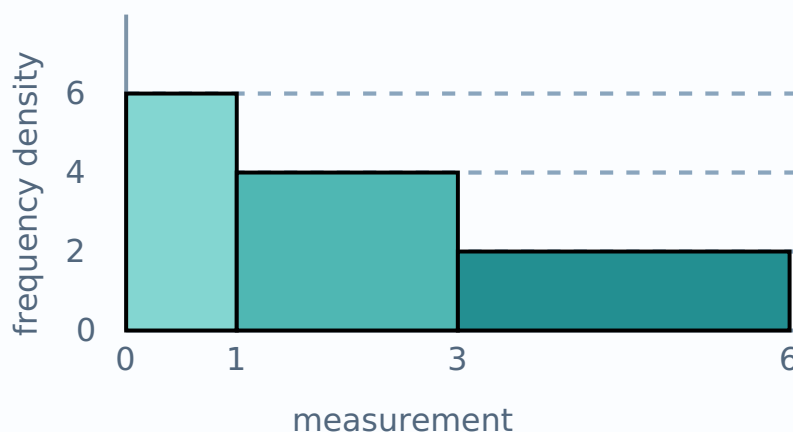
Worked solution

Answer A

In right triangle QRS , $QR/RS = 4/3$, so $QR = 28$. The horizontal distance from P to the vertical through Q is $21/\tan 30^\circ = 21\sqrt{3}$. Therefore $PS = 28 + 21\sqrt{3}$.

Question 10

A histogram represents exactly 20 measurements. Its class intervals are $0 \leq x < 1$, $1 \leq x < 3$, and $3 \leq x \leq 6$, where x is the measurement. The frequency densities are shown below.



The unequal widths are part of the histogram.

The mean of all 20 measurements is exactly 1.8. What is the greatest possible number of measurements that are at least 5?

A 3

B 5

C 8

D 4

E 6

Worked solution

Answer B

Frequency equals class width multiplied by frequency density, so the three frequencies are $1 \times 6 = 6$, $2 \times 4 = 8$, and $3 \times 2 = 6$. The total of all measurements is $20(1.8) = 36$.

Only the six observations in $[3, 6]$ can be at least 5. If all six were, the total would be at least $6(5) + 8(1) + 6(0) = 38$, which is impossible. Five are achievable: take five values 5, one value 3, eight values 1, and six values 0. These respect the stated class boundaries and sum to 36. The maximum number is therefore 5.

Question 11

Two fair six-sided dice are rolled. Given that at least one die shows 4, what is the probability that the sum of the scores is at least 9?

A $\frac{1}{3}$

B $\frac{2}{5}$

C $\frac{4}{11}$

D $\frac{5}{12}$

E $\frac{1}{2}$

Worked solution

Answer C

Given that at least one die shows 4, there are $6 + 6 - 1 = 11$ equally likely ordered outcomes. The outcomes with sum at least 9 are (4, 5), (4, 6), (5, 4), (6, 4). Therefore the probability is $4/11$.

Question 12

For positive integers n , define $a_n = n^2 + 3n + 2$. How many of the first 120 terms are divisible by 12?

A 32

B 36

C 44

D 40

E 48

Worked solution

Answer D

Factor $a_n = (n+1)(n+2)$. Divisibility by 12 requires divisibility by both 3 and 4.

For the factor 4, the even member of the two consecutive integers must be a multiple of 4. Thus n leaves remainder 2 or 3 when divided by 4.

For the factor 3, n must leave remainder 1 or 2 when divided by 3. In each block of 12 consecutive values of n , both requirements hold at remainders 2, 7, 10, 11.

There are $120/12 = 10$ complete blocks, each with 4 successes, giving 40 terms.

Question 13

The graph of $y = g(x)$ is made from straight-line segments joining, in order, the points $(-4, 3)$, $(-1, -3)$, $(2, 1)$, $(5, -2)$, $(7, 2)$. What is the sum of the x -coordinates of all its intersections with the x -axis?

A $\frac{29}{4}$

B $\frac{31}{4}$

C 8

D $\frac{33}{4}$

E $\frac{35}{4}$

Worked solution

Answer B

On the first segment, the heights 3 and -3 put the crossing halfway between $x = -4$ and $x = -1$, at $-\frac{5}{2}$. On the second segment, the height rises from -3 to 1, so zero is reached three quarters of the way from -1 to 2: $-1 + \frac{3}{4}(3) = \frac{5}{4}$.

On the third segment, the height falls from 1 to -2 , so the crossing is one third of the way from 2 to 5, at 3. The final segment crosses halfway from 5 to 7, at 6. Their sum is $-\frac{5}{2} + \frac{5}{4} + 3 + 6 = \frac{31}{4}$.

Question 14

For a real parameter a , consider $x^2 - 2(a+1)x + a^2 - 1 = 0$. For which values of a does this equation have one positive root and one negative root?

A $a \leq -1$

B $-1 \leq a < 0$

C $-1 < a < 1$

D $0 < a \leq 1$

E $a < -1$ or $a > 1$

Worked solution

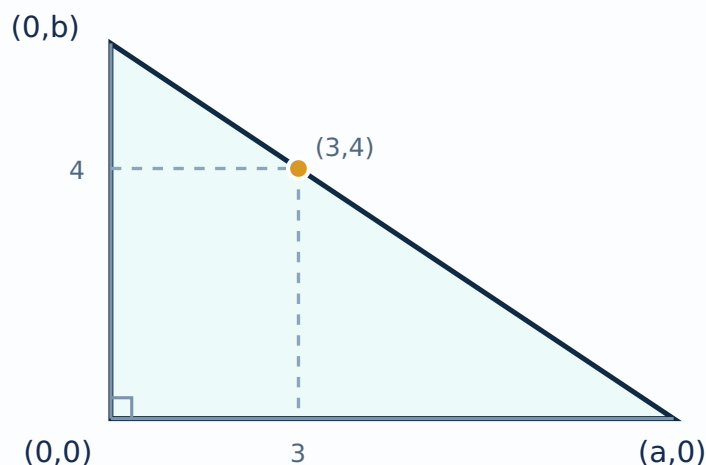
Answer C

Writing a monic quadratic as $(x-r)(x-s)$ shows that its constant term is the product rs . Here the product is $a^2 - 1$. Opposite signs require a negative product, so $a^2 - 1 < 0$, giving $-1 < a < 1$.

This condition is also sufficient: the upward-opening quadratic is negative at $x = 0$ and positive for sufficiently large positive or negative x , so it crosses the axis once on each side of zero. The endpoints $a = \pm 1$ are excluded because one root is then zero.

Question 15

A right triangle has vertices $(0,0)$, $(a,0)$, $(0,b)$, where $a, b > 0$. The point $(3,4)$ lies on its hypotenuse. What is the minimum possible area of the triangle?



Schematic diagram; not to scale.

A 18

B 20

C 32

D 28

E 24

Worked solution

Answer E

The line through $(a,0)$ and $(0,b)$ has equation $y = b - \frac{b}{a}x$, equivalently $\frac{x}{a} + \frac{y}{b} = 1$. Since $(3,4)$ lies on it, $\frac{3}{a} + \frac{4}{b} = 1$. Set $u = \frac{3}{a}$ and $v = \frac{4}{b}$, so $u + v = 1$ and $ab = \frac{12}{uv}$.

Because $(u - v)^2 \geq 0$, we have $4uv \leq (u + v)^2 = 1$. Therefore $ab \geq 48$, and the area is at least $\frac{1}{2}(48) = 24$. Equality occurs at $u = v = \frac{1}{2}$, giving $a = 6$ and $b = 8$, so this minimum is attained.

Question 16

A sequence is defined by $a_1 = 1$ and $a_{n+1} = 3a_n + 2$. What is the remainder when a_{2026} is divided by 7?

A 4

B 2

C 1

D 5

E 6

Worked solution

Answer A

The successive remainders on division by 7 are 1, 5, 3, 4, 0, 2, then repeat. Since $2026 = 6 \cdot 337 + 4$, the required remainder is the fourth one in the cycle, namely 4.

Question 17

Points $A = (-4, 7)$, $B = (5, -2)$ and $C = (3, 6)$ are given. Point P lies on AB such that $AP : PB = 2 : 1$. Point Q is the reflection of C in P , meaning a rotation of 180° about P . What is $2x_Q - y_Q$?

A -6**B** -2**C** 2**D** 6**E** 10

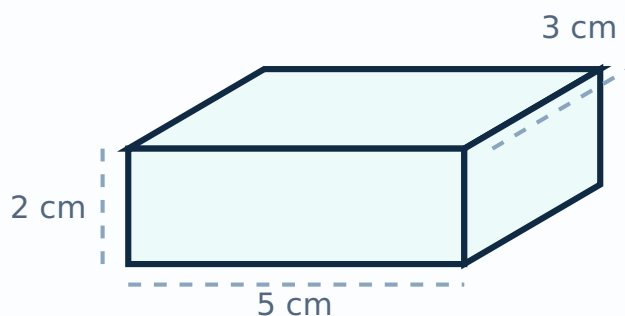
Worked solution

Answer D

$P = A + \frac{2}{3}(B - A) = (-4, 7) + \frac{2}{3}(9, -9) = (2, 1)$. Since P is the midpoint of CQ ,
 $Q = 2P - C = (1, -4)$. Thus $2x_Q - y_Q = 2 + 4 = 6$.

Question 18

A rectangular metal block measures $2\text{ cm} \times 3\text{ cm} \times 5\text{ cm}$ and has mass 270 g . What is its density in kg m^{-3} ?



Schematic block; dimensions are not drawn to scale.

A 90

B 900

C 9000

D 27000

E 90000

Worked solution

Answer C

The block has volume $2 \times 3 \times 5 = 30\text{ cm}^3 = 3 \times 10^{-5}\text{ m}^3$. Its mass is $270\text{ g} = 0.270\text{ kg}$.
Therefore its density is $\frac{0.270}{3 \times 10^{-5}} = 9000\text{ kg m}^{-3}$.

Question 19

Let O, A, B, C be the vertices of a parallelogram, with O as the origin and $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ as adjacent sides. Thus C is the fourth vertex. Point P lies on AC with $AP : PC = 1 : 2$, and Q is the midpoint of OB . Which expression is $\overrightarrow{PQ}$?

A $\mathbf{a} + \frac{1}{6}\mathbf{b}$

B $-\mathbf{a} + \frac{1}{6}\mathbf{b}$

C $-\mathbf{a} - \frac{1}{6}\mathbf{b}$

D $\mathbf{a} - \frac{1}{6}\mathbf{b}$

E $-\mathbf{a} + \frac{1}{3}\mathbf{b}$

Worked solution

Answer B

Because OA and OB are adjacent sides, $\overrightarrow{OC} = \mathbf{a} + \mathbf{b}$, so $P = \mathbf{a} + \frac{1}{3}\mathbf{b}$. Also $Q = \frac{1}{2}\mathbf{b}$.
Therefore $\overrightarrow{PQ} = Q - P = -\mathbf{a} + \frac{1}{6}\mathbf{b}$.

Question 20

The line $y = 2x - 1$ is tangent to the graph $y = x^2 - 2kx + k + 3$, where k is real. What is the sum of the two possible values of k ?

A -3

B -2

C 3

D 1

E -1

Worked solution

Answer E

At an intersection, $x^2 - 2(k+1)x + k + 4 = 0$. Completing the square gives $(x - k - 1)^2 = k^2 + k - 3$. A tangent has exactly one point of contact, so the right-hand side must be zero.

Thus $k^2 + k - 3 = 0$, giving $k = \frac{-1+\sqrt{13}}{2}$ or $k = \frac{-1-\sqrt{13}}{2}$. The sum is -1 .

Tip: a repeated root can be recognised by completing the square; differentiation is not required.

Question 21

A quantity P is directly proportional to x^2 and inversely proportional to y . If x is increased by 20% and y is decreased by 20%, by what factor does P change?

A $\frac{4}{5}$

B $\frac{6}{5}$

C $\frac{9}{5}$

D 2

E $\frac{12}{5}$

Worked solution

Answer C

Write $P = c\frac{x^2}{y}$ for some constant c . After the changes, x is multiplied by $6/5$ and y by $4/5$, so P is multiplied by $\frac{(6/5)^2}{4/5} = \frac{9}{5}$.

Question 22

Six cards are numbered 1, 2, 3, 4, 5, 6. Two different cards are selected, with every pair equally likely.

Given that the product of their numbers is even, what is the probability that their sum is greater than 7?

A $\frac{1}{3}$

B $\frac{1}{5}$

C $\frac{5}{12}$

D $\frac{1}{2}$

E $\frac{3}{5}$

Worked solution

Answer C

There are $6 \times 5/2 = 15$ unordered pairs. The product is odd only when both cards are odd. The three odd cards form 3 pairs, so $15 - 3 = 12$ pairs satisfy the condition.

The eligible pairs whose sum exceeds 7 are

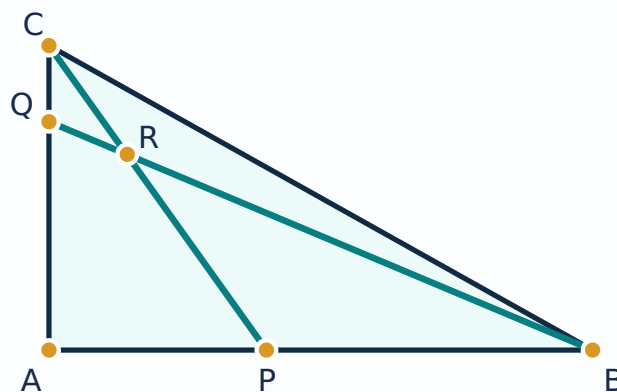
$$(2, 6), (3, 6), (4, 5), (4, 6), (5, 6).$$

The pair (3, 5) has a large enough sum but has an odd product. The required probability is $5/12$.

Useful approach: Apply the given condition to both the denominator and the favourable outcomes.

Question 23

In triangle ABC , let $A = (0, 0)$, $B = (10, 0)$, and $C = (0, 8)$. Point P lies on AB with $AP : PB = 2 : 3$, and point Q lies on AC with $AQ : QC = 3 : 1$. Lines CP and BQ meet at R . What is the x -coordinate of R ?



Schematic diagram; not to scale.

A 1

B $\frac{4}{3}$

C $\frac{5}{3}$

D $\frac{10}{7}$

E 2

Worked solution

Answer D

$P = (4, 0)$, $Q = (0, 6)$. The two lines are $CP : y = 8 - 2x$ and $BQ : y = 6 - \frac{3}{5}x$. Equating gives $2 = \frac{7}{5}x$, hence $x = 10/7$.

Question 24

Box A is chosen with probability $\frac{2}{3}$, and box B with probability $\frac{1}{3}$. Box A contains 2 red and 3 blue counters; box B contains 4 red and 1 blue counter. Two counters are drawn without replacement from the chosen box. Given that the two counters have different colours, what is the probability that box A was chosen?

A $\frac{1}{2}$

B $\frac{2}{3}$

C $\frac{7}{10}$

D $\frac{3}{4}$

E $\frac{4}{5}$

Worked solution

Answer D

Let D mean that the two drawn counters have different colours. From box A , the red-blue and blue-red branches give $P(D | A) = 2(\frac{2}{5})(\frac{3}{4}) = \frac{3}{5}$. For B , $P(D | B) = 2(\frac{4}{5})(\frac{1}{4}) = \frac{2}{5}$.

The probability of choosing A and then drawing different colours is $\frac{2}{3} \cdot \frac{3}{5} = \frac{2}{5}$. The probability of different colours overall is $\frac{2}{5} + \frac{1}{3} \cdot \frac{2}{5} = \frac{8}{15}$. Of these different-colour outcomes, the fraction originating from A is $\frac{2/5}{8/15} = \frac{3}{4}$.

Question 25

How many ordered pairs of positive integers (a, b) satisfy $ab = 360$, have highest common factor 1, and have $a + b$ a multiple of 7?

A 2

B 0

C 6

D 8

E 4

Worked solution

Answer E

Since $360 = 2^3 \cdot 3^2 \cdot 5$ and a, b are coprime, the prime powers 8, 9, 5 must be assigned wholly to one factor. The possible unordered splits are $1 + 360, 8 + 45, 9 + 40, 72 + 5$; only $9 + 40 = 49$ and $72 + 5 = 77$ are multiples of 7. Both orders count, giving 4.

Question 26

Data set A has mean 14, and data set B has mean 20. The mean of all observations in the two data sets combined is 16. An observation of value 10 from A is exchanged with an observation of value 34 from B . The two data sets then have the same mean. How many observations are there altogether?

A 6**B** 12**C** 18**D** 24**E** 36

Worked solution

Answer C

Exchanging the two observations changes neither the combined total nor the size of either data set. The combined mean remains 16, so the two equal new means must both be 16.

The total in A increases by $34 - 10 = 24$, while its mean increases by $16 - 14 = 2$. Therefore A contains $24/2 = 12$ observations. The total in B decreases by 24, while its mean decreases by $20 - 16 = 4$, so B contains $24/4 = 6$ observations. Altogether there are $12 + 6 = 18$.

Trap: averaging the two original means without accounting for the different data set sizes.

Question 27

Two geometrically similar solid objects are made from the same uniform material. Their total surface areas are in the ratio $4 : 9$, smaller to larger. The mass of the larger object is 95 g greater than the mass of the smaller object. What is the mass of the smaller object?

A 40 g

B 20 g

C 76 g

D 95 g

E 190 g

Worked solution

Answer A

The surface area ratio $4 : 9$ gives a length ratio $2 : 3$. The volume ratio is therefore $2^3 : 3^3 = 8 : 27$. Because the objects are made from the same uniform material, their masses also have ratio $8 : 27$.

The difference of $27 - 8 = 19$ ratio parts is 95 g, so one part is 5 g. The smaller mass is $8 \times 5 = 40$ g.

Trap: applying the surface area ratio directly to mass gives 76 g; using the length ratio gives 190 g.

MODULE 02

Mathematics 2

QUESTIONS & WORKED SOLUTIONS

Maths 2 + Biology pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

Using this module

Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

The curve $y = \frac{(x+2)(x-1)}{\sqrt{x}}$ is defined for $x > 0$. What is the gradient of the normal to the curve at $x = 4$?

A $-\frac{27}{8}$

B $-\frac{8}{25}$

C $\frac{8}{27}$

D $-\frac{8}{27}$

E $\frac{27}{8}$

Worked solution

Answer D

Expand the numerator and divide each term by $\sqrt{x}$:

$$y = x^{3/2} + x^{1/2} - 2x^{-1/2}.$$

Differentiate the powers separately:

$$\frac{dy}{dx} = \frac{3}{2}x^{1/2} + \frac{1}{2}x^{-1/2} + x^{-3/2}.$$

At $x = 4$, the tangent gradient is $3 + \frac{1}{4} + \frac{1}{8} = \frac{27}{8}$. A normal is perpendicular to the tangent, so its gradient is the negative reciprocal, $-\frac{8}{27}$.

Question 02

A sequence is defined by $u_1 = 3$ and $u_{n+1} = \frac{3u_n - 1}{u_n + 1}$. What is $\prod_{n=1}^{20} u_n$?

A 21

B 42

C 120

D 220

E 231

Worked solution

Answer E

The first few terms are $3, 2, \frac{5}{3}$, suggesting $u_n = \frac{n+2}{n}$. This formula holds at $n = 1$. If it holds at a particular n , the recurrence gives

$$u_{n+1} = \frac{3(n+2)/n - 1}{(n+2)/n + 1} = \frac{2n+6}{2n+2} = \frac{n+3}{n+1},$$

so it holds at the next term as well. Consequently,

$$\prod_{n=1}^{20} u_n = \frac{3}{1} \cdot \frac{4}{2} \cdot \frac{5}{3} \cdots \frac{22}{20} = \frac{21 \cdot 22}{1 \cdot 2} = \boxed{231}.$$

All factors from 3 to 20 cancel between numerator and denominator.

Question 03

For $0 \leq x \leq 2\pi$, with the original expression defined, $\frac{\sin^2 x}{1 - \cos x} = 2 \sin x$. What is the sum of $\cos x$ over all distinct real solutions?

A $-\frac{2}{5}$

B $-\frac{1}{2}$

C 0

D $\frac{2}{5}$

E $\frac{1}{2}$

Worked solution

Answer A

The denominator excludes $x = 0, 2\pi$. For all allowed solutions, $1 + \cos x = 2 \sin x$, so $\sin x \geq 0$ and hence $0 < x \leq \pi$. Squaring and using $\sin^2 x = 1 - \cos^2 x$ gives $5 \cos^2 x + 2 \cos x - 3 = 0$. Thus $\cos x = 3/5$ or -1 , both satisfying the original equation. The sum is $-1 + 3/5 = -2/5$.

Question 04

A continuous function f satisfies $f(x) - f(-x) = 2x^3 - 6x$. What is $\int_{-2}^2 xf(x) dx$?

A $-\frac{16}{3}$

B $-\frac{16}{5}$

C $-\frac{4}{5}$

D $\frac{16}{5}$

E $\frac{16}{3}$

Worked solution

Answer B

Pair the integrand at x and $-x$: $xf(x) + (-x)f(-x) = x[f(x) - f(-x)] = 2x^4 - 6x^2$.
Therefore the integral is $\int_0^2 (2x^4 - 6x^2) dx = 64/5 - 16 = -16/5$.

Question 05

For a positive integer n , the coefficient of x^4 in $(1-x)^n(1+x)^{n+2}$ is 27. What is the coefficient of x^6 ?

A -48

B -36

C -24

D 24

E 48

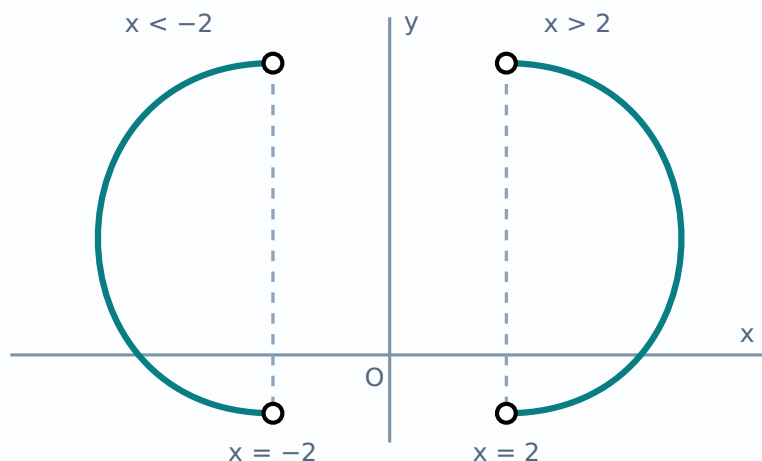
Worked solution

Answer A

Rewrite the product as $(1+x)^2(1-x^2)^n$. The coefficient of x^4 is $\binom{n}{2} - n = n(n-3)/2$, so $n = 9$. The coefficient of x^6 is $-\binom{9}{3} + \binom{9}{2} = -84 + 36 = -48$.

Question 06

The equation $\log_2(5 - y) + \log_2(y + 1) = 2\log_2(|x| - 2)$ defines a locus. What is the total length of the locus?



Open circles mark excluded endpoints; schematic, not to scale.

A $\frac{3\pi}{2}$

B 2π

C 4π

D 6π

E 9π

Worked solution

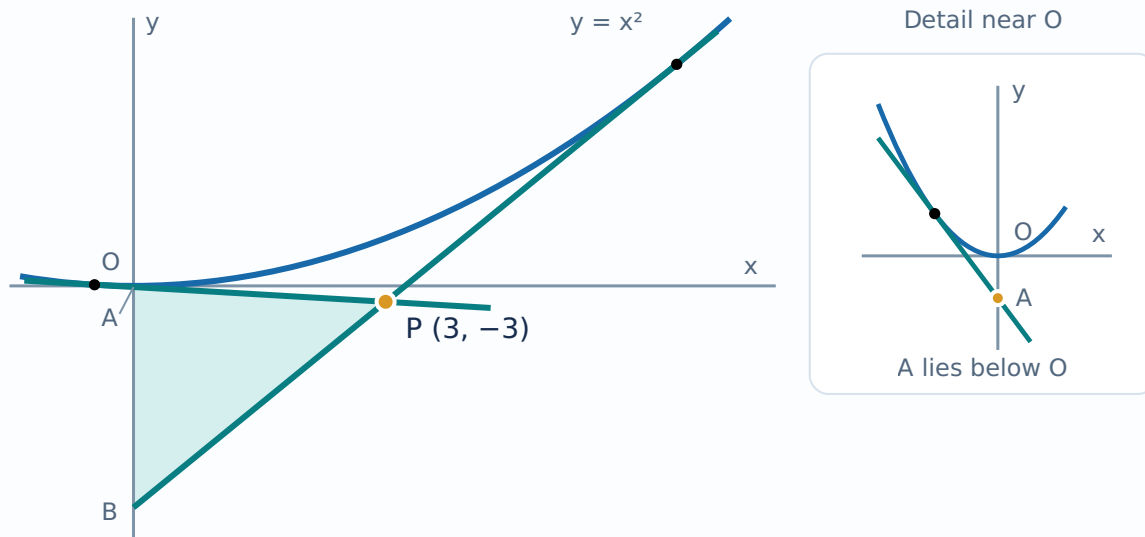
Answer D

The logarithms require $-1 < y < 5$ and $|x| > 2$. Combining and exponentiating gives $(y - 2)^2 + (|x| - 2)^2 = 9$. The restrictions select two open semicircular arcs, each of radius 3, so the total length is $2(3\pi) = 6\pi$.

Trap: keeping the forbidden halves of the two circles after completing the square.

Question 07

Two tangents are drawn from $(3, -3)$ to the parabola $y = x^2$. What is the area of the region enclosed by the two tangents and the y -axis?



The tangents and y -axis bound the shaded triangle. Axes use different scales; the inset enlarges the origin.

A $36\sqrt{3}$

B $18\sqrt{3}$

C $12\sqrt{3}$

D 36

E $72\sqrt{3}$

Worked solution

Answer A

A tangent through $(3, -3)$ with gradient m is $y + 3 = m(x - 3)$. Intersections with $y = x^2$ give $x^2 - mx + 3m + 3 = 0$; tangency gives $m^2 - 12m - 12 = 0$, so $m = 6 \pm 4\sqrt{3}$. The two y -axis intercepts differ by $3(8\sqrt{3}) = 24\sqrt{3}$. With horizontal distance 3 from $(3, -3)$ to the y -axis, the area is $\frac{1}{2}(24\sqrt{3})(3) = 36\sqrt{3}$.

Question 08

Let $f(x) = |x^2 - 4x + 3|$. For which positive value of k does $f(f(x)) = k$ have exactly 8 distinct real solutions?

A $\frac{3}{2}$

B $\frac{1}{2}$

C 1

D 2

E 3

Worked solution

Answer C

Write $f(x) = |(x - 2)^2 - 1|$. For $t \geq 0$, solve $f(x) = t$ by using $(x - 2)^2 = 1 + t$ or $(x - 2)^2 = 1 - t$. There are two solutions for $t = 0$, four for $0 < t < 1$, three for $t = 1$, and two for $t > 1$.

Now put $y = f(x)$, so $y \geq 0$. For $k = 1$, solving $f(y) = 1$ gives the three distinct nonnegative values

$$y = 2 - \sqrt{2}, \quad 2, \quad 2 + \sqrt{2}.$$

The first lies between 0 and 1, so it contributes four possible values of x . Each of the other two exceeds 1, so each contributes two. There are $4 + 2 + 2 = 8$ distinct solutions.

For the other listed values $k = \frac{3}{2}, \frac{1}{2}, 2, 3$, the same counting gives 6, 10, 6, 4 solutions respectively. Hence the required value is 1.

Question 09

A sequence has n th term $u_n = 6n - 1$. From the first 15 terms, every term whose position is a multiple of 3 is removed. What is the sum of the terms that remain?

A 265

B 435

C 440

D 450

E 705

Worked solution

Answer C

The first and fifteenth terms are 5 and 89, so the sum of all fifteen terms is

$$S = \frac{15}{2}(5 + 89) = 705.$$

The removed terms are $u_3, u_6, u_9, u_{12}, u_{15}$, namely 17, 35, 53, 71, 89. They form an arithmetic progression with sum

$$R = \frac{5}{2}(17 + 89) = 265.$$

The required sum is $705 - 265 = \boxed{440}$. The multiples of 3 refer to the term positions, not the values of the terms.

Question 10

For $0 < x < \frac{\pi}{2}$, $\sin^2 x + \sin^4 x + \sin^6 x + \cdots = 3 \cos^2 x$. What is $\tan^2 x$?

A $\frac{\sqrt{13} + 1}{2}$

B $\frac{\sqrt{13} - 1}{2}$

C $\frac{\sqrt{13} - 3}{2}$

D $\frac{\sqrt{13} + 3}{2}$

E 3

Worked solution

Answer B

The series is $\frac{\sin^2 x}{1 - \sin^2 x} = \tan^2 x$. Let $t = \tan^2 x$. Since $\cos^2 x = 1/(1 + t)$, $t = 3/(1 + t)$, so $t^2 + t - 3 = 0$. The positive root is $(\sqrt{13} - 1)/2$.

Question 11

For $0 \leq x \leq 4$, consider the region between the curve $y = x^2 - 4x + 3$ and the x -axis. What fraction of the total area of this region lies below the x -axis?

A $\frac{1}{4}$

B $\frac{1}{3}$

C $\frac{1}{2}$

D $\frac{2}{3}$

E $\frac{3}{4}$

Worked solution

Answer B

Factorise $x^2 - 4x + 3 = (x - 1)(x - 3)$. The curve is below the axis for $1 < x < 3$, and above it on the two remaining intervals.

An antiderivative is $F(x) = \frac{x^3}{3} - 2x^2 + 3x$. Its values at 0, 1, 3, 4 are $0, \frac{4}{3}, 0, \frac{4}{3}$, respectively. Thus each of the three separate regions has area $\frac{4}{3}$:

$$A_{\text{below}} = - \int_1^3 (x^2 - 4x + 3) dx = \frac{4}{3}, \quad A_{\text{total}} = 4.$$

The fraction below the axis is $\frac{4/3}{4} = \boxed{\frac{1}{3}}$. The signed integral over $[0, 4]$ is not the total area.

Question 12

The equation $3^{2x} - a3^x + 9 = 0$ has distinct real roots u, v satisfying $u^2 + v^2 = 10$. What is a ?

A $\frac{28}{3}$

B $\frac{244}{9}$

C 26

D $\frac{82}{3}$

E 28

Worked solution

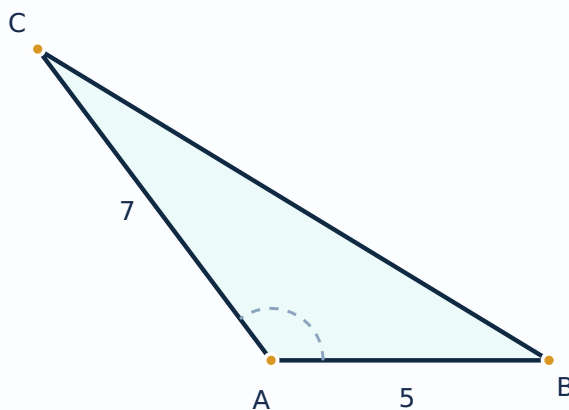
Answer D

Put $t = 3^x$. The quadratic in t has roots $3^u, 3^v$, whose product is 9, so $u + v = 2$. From $u^2 + v^2 = 10$, $uv = -3$, hence $\{u, v\} = \{-1, 3\}$. Therefore $a = 3^{-1} + 3^3 = 82/3$.

Trap: applying Vieta directly to the exponents instead of to $3^u, 3^v$.

Question 13

Triangle ABC has $AB = 5$, $AC = 7$, and area 14. Angle BAC is obtuse. What is the length of BC ?



Angle BAC is obtuse. No measurement from the diagram is required.

A $4\sqrt{2}$

B $\sqrt{46}$

C $\sqrt{74}$

D $\sqrt{102}$

E $2\sqrt{29}$

Worked solution

Answer E

Let $\theta = \angle BAC$. The area formula gives

$$14 = \frac{1}{2}(5)(7) \sin \theta, \quad \sin \theta = \frac{4}{5}.$$

Since the angle is obtuse, its cosine is negative. Using $\sin^2 \theta + \cos^2 \theta = 1$ gives $\cos \theta = -\frac{3}{5}$.

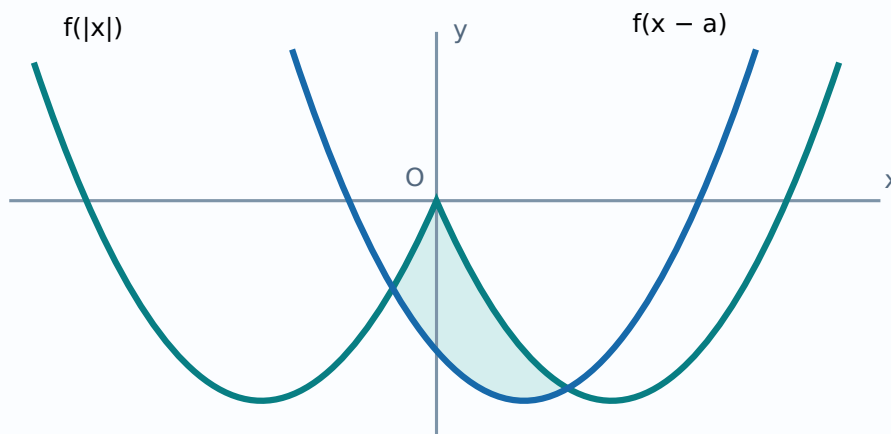
Now apply the cosine rule:

$$BC^2 = 5^2 + 7^2 - 2(5)(7) \left(-\frac{3}{5}\right) = 74 + 42 = 116.$$

Therefore $BC = \sqrt{116} = \boxed{2\sqrt{29}}$. Choosing the positive value of the cosine would give the acute-angle triangle instead.

Question 14

Let $f(x) = x^2 - 4x$ and suppose $-4 < a < 0$. The curves $y = f(|x|)$ and $y = f(x - a)$ enclose a region of area 3. What is the sum of the squares of all possible values of a ?



Representative configuration for $-4 < a < 0$; the shaded region lies between the curves.

A 6

B 16

C 12

D 8

E 10

Worked solution

Answer E

The difference $f(|x|) - f(x - a)$ is linear on each side of 0, with zeros at $x = a/2$ and $x = (a + 4)/2$. For $-4 < a < 0$, this difference is positive between its zeros. The region under the graph of the difference, above the x -axis, is a triangle of base 2 and height $-a(a + 4)$. Therefore the area between the original curves is $-a(a + 4)$. Setting this to 3 gives $a = -1, -3$, and the required sum of squares is $1 + 9 = 10$.

Question 15

A monic polynomial P of degree 4 has remainder $2x + 1$ when divided by $x^2 - 1$, and remainder $x + 7$ when divided by $x^2 - 4$. What is the least value of $P(x) + P(-x)$ for real x ?

A $\frac{3}{4}$

B 6

C $\frac{3}{2}$

D 3

E $\frac{9}{2}$

Worked solution

Answer C

Write $P(x) = x^4 + bx^3 + cx^2 + dx + e$. When reducing the polynomial using $x^2 = 1$, its constant part is $1 + c + e$. The first remainder therefore gives $1 + c + e = 1$, or $c + e = 0$.

Using $x^2 = 4$, the constant part is $16 + 4c + e$. The second remainder gives $16 + 4c + e = 7$, or $4c + e = -9$. Subtracting yields $c = -3$, and hence $e = 3$.

The odd powers cancel in the required sum:

$$P(x) + P(-x) = 2(x^4 - 3x^2 + 3) = 2\left(x^2 - \frac{3}{2}\right)^2 + \frac{3}{2}.$$

The squared term is zero at the real values $x = \pm\sqrt{\frac{3}{2}}$. Thus the least value is $\boxed{\frac{3}{2}}$.

Question 16

What is the total length of the intervals within $0 \leq x \leq 2\pi$ for which $2\sin^2 x + 3\cos x > 0$?

A $\frac{\pi}{3}$

B $\frac{2\pi}{3}$

C π

D $\frac{4\pi}{3}$

E $\frac{5\pi}{3}$

Worked solution

Answer D

Use $\sin^2 x = 1 - \cos^2 x$. The inequality becomes

$$2 + 3\cos x - 2\cos^2 x = (2 - \cos x)(2\cos x + 1) > 0.$$

For every real x , $-1 \leq \cos x \leq 1$, so $2 - \cos x$ is positive. Therefore the condition is simply $\cos x > -\frac{1}{2}$.

Within the stated domain, this holds on $[0, \frac{2\pi}{3})$ and $(\frac{4\pi}{3}, 2\pi]$. Their lengths add to

$\frac{2\pi}{3} + \frac{2\pi}{3} = \boxed{\frac{4\pi}{3}}$. Excluding isolated endpoints does not change an interval's length.

Question 17

A function f , defined for every real number, has exactly four distinct real zeros. The equation $f(x + 1/x) = 0$, with $x \neq 0$, has exactly five distinct real solutions. How many of the zeros of f must lie strictly between -2 and 2 ?

A 2**B** 3**C** 0**D** 4**E** 1

Worked solution

Answer E

For a zero r of f , the equation $x + 1/x = r$ becomes $x^2 - rx + 1 = 0$. Its discriminant is $r^2 - 4$, so this zero contributes zero, one, or two real solutions according as $|r| < 2$, $|r| = 2$, or $|r| > 2$. None of these quadratic roots is zero.

Let h be the number of zeros of f outside $[-2, 2]$, and let b be the number at the boundary values -2 and 2 . Different zeros give disjoint sets of solutions, so $2h + b = 5$, with $0 \leq b \leq 2$.

The left side is odd only when $b = 1$. Hence $h = 2$, leaving $4 - 2 - 1 = \boxed{1}$ zero strictly between -2 and 2 .

Question 18

Consider the claim: every real polynomial f with $f(0) < 0$, $f(2) > 0$, and $f'(0) > 0$ has exactly one zero in $(0, 2)$. Which polynomial is a counterexample?

A $-4x^3 + 12x^2 - 11x + 3$

B $x^3 + x - 1$

C $4x^3 - 12x^2 + 11x - 3$

D $x^3 - x - 1$

E $4x^3 - 12x^2 + 8x$

Worked solution

Answer C

Option C factorises as

$$4x^3 - 12x^2 + 11x - 3 = (2x - 1)(x - 1)(2x - 3).$$

It has three distinct zeros, $\frac{1}{2}$, 1 , $\frac{3}{2}$, all in $(0, 2)$. It also satisfies all three hypotheses: $f(0) = -3 < 0$, $f(2) = 3 > 0$, and $f'(0) = 11 > 0$. It is therefore a counterexample.

A and E fail the requirement $f(0) < 0$, while D fails $f'(0) > 0$. B does satisfy the hypotheses, but its derivative $3x^2 + 1$ is positive everywhere; it has exactly one zero in the interval, so it does not disprove the claim.

Question 19

Let n be a positive integer. Which condition is necessary and sufficient for n^3 to be divisible by 72 while n^2 is not divisible by 72?

A $n = 6k + 3$ for some integer $k \geq 0$

B n is divisible by 6

C $n = 12k + 6$ for some integer $k \geq 0$

D n is divisible by 12

E $n = 24k + 12$ for some integer $k \geq 0$

Worked solution

Answer C

Factorise $72 = 2^3 \cdot 3^2$. For 72 to divide n^3 , the integer n must contain at least one factor of 2 and at least one factor of 3. Equivalently, 6 divides n .

Then n^2 already contains the required 3^2 . It fails to be divisible by 72 precisely when it lacks 2^3 , which means that n contains exactly one factor of 2.

Thus n must be divisible by 6 but not by 12. This is exactly $n = 12k + 6$ for an integer $k \geq 0$. Every integer of this form also satisfies both original divisibility conditions.

Question 20

Let $f(x) = x^5 - 5ax^3 + 5x$. For which values of a is $f'(x) > 0$ for every real x ?

A $a \leq -\frac{2}{3}$

B $a < \frac{2}{3}$

C $a \leq \frac{2}{3}$

D $a > \frac{2}{3}$

E all real a

Worked solution

Answer B

Differentiate and put $t = x^2 \geq 0$:

$$f'(x) = 5(x^4 - 3ax^2 + 1) = 5(t^2 - 3at + 1).$$

If $a \leq 0$, all the terms in $t^2 - 3at + 1$ are nonnegative and the constant term is 1, so the expression is positive.

If $a > 0$, the quadratic has its minimum for $t \geq 0$ at $t = \frac{3a}{2}$. That minimum is $1 - \frac{9a^2}{4}$, which is strictly positive exactly when $a < \frac{2}{3}$.

Combining the cases gives $a < \frac{2}{3}$. At $a = \frac{2}{3}$, the derivative equals $5(x^2 - 1)^2$, which is zero at $x = \pm 1$, so equality is excluded.

Question 21

A differentiable function f satisfies $f'(x) = 6x - 12$ for every real x , and $\int_0^2 f(x) dx = 0$. What is $f(3)$?

A -9

B -1

C 1

D 7

E 15

Worked solution

Answer B

Integrating the derivative gives $f(x) = 3x^2 - 12x + C$, where C is a constant. Use the definite-integral condition to determine it:

$$0 = \int_0^2 (3x^2 - 12x + C) dx = \left[x^3 - 6x^2 + Cx \right]_0^2 = 8 - 24 + 2C.$$

Thus $C = 8$, and $f(3) = 27 - 36 + 8 = \boxed{-1}$. A zero definite integral does not mean that the function itself is zero at either endpoint.

Question 22

A real polynomial f of degree exactly 4 has exactly three distinct stationary points at $x = a, b, c$, where $a < b < c$. It satisfies $f(a) = f(c) = 0$ and $f(b) > 0$. Consider: I. $b = (a + c)/2$. II. $f(x) \geq 0$ for every real x . III. For every real $d > 0$, $f(x) = d$ has exactly four distinct real solutions. Which statements must be true?

A II only

B I and III only

C I only

D I, II and III

E I and II only

Worked solution

Answer E

A stationary point at which the polynomial equals zero is a repeated root: both f and f' vanish there. Thus a and c are each roots of multiplicity at least two. Since the degree is exactly four,

$$f(x) = K(x - a)^2(x - c)^2$$

for a nonzero constant K . Because $f(b) > 0$ and b differs from a, c , we have $K > 0$. Hence $f(x) \geq 0$ everywhere, proving II.

Expanding the two quadratic factors and differentiating gives the factorisation

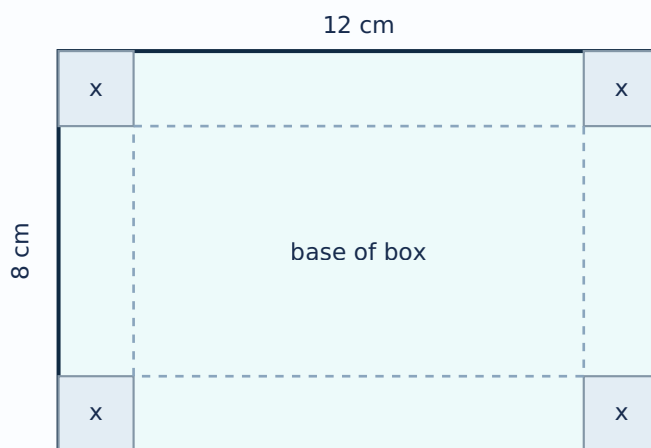
$$f'(x) = 2K(x - a)(x - c)(2x - a - c).$$

The third stationary point is therefore $b = (a + c)/2$, proving I. The graph has minima of zero at a, c and a maximum between them at b . For $d > f(b)$, only the two outer branches meet $y = d$, giving two solutions rather than four. III is false.

The correct selection is I and II only.

Question 23

A rectangular sheet measures 12 cm by 8 cm. A square of side x cm is cut from each corner, and the remaining sides are folded up to make an open box. For $0 < x < 4$, which value of x gives the greatest volume?



Shaded corner squares are removed. Dashed lines show the folds; x is illustrative.

A $\frac{10 - 2\sqrt{7}}{3}$

B $\frac{5 - \sqrt{7}}{3}$

C 2

D $\frac{10 - \sqrt{7}}{3}$

E $\frac{20 - 4\sqrt{7}}{3}$

Worked solution

Answer A

The box has height x , length $12 - 2x$, and width $8 - 2x$. Its volume is

$$V(x) = x(12 - 2x)(8 - 2x) = 96x - 40x^2 + 4x^3.$$

Differentiate and set the derivative equal to zero:

$$V'(x) = 96 - 80x + 12x^2 = 0 \iff 3x^2 - 20x + 24 = 0.$$

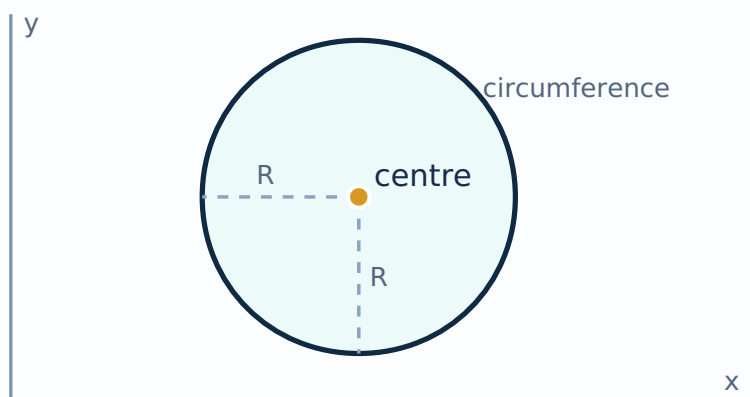
The roots are $x = \frac{10 \pm 2\sqrt{7}}{3}$. Since $2 < \sqrt{7} < 3$, the larger root exceeds 4, and only the smaller root lies in the permitted interval.

The upward-opening quadratic $V'(x)$ is positive before the smaller root and negative between the two roots. Hence the volume increases and then decreases on $(0, 4)$. Its

greatest value occurs at $x = \frac{10 - 2\sqrt{7}}{3}$.

Question 24

For real a, b, c , which condition is necessary and sufficient for $x^2 + y^2 + 2ax + 2by + c = 0$ to describe a nondegenerate circle whose entire circumference lies in the open first quadrant $x > 0, y > 0$?



Strict containment means the circle cannot touch either coordinate axis; schematic, not to scale.

A $a < 0, b < 0, \max(a^2, b^2) < c < a^2 + b^2$

B $a < 0, b < 0, \max(a^2, b^2) \leq c < a^2 + b^2$

C $a < 0, b < 0, 0 < c < a^2 + b^2$

D $a < 0, b < 0, c > a^2 + b^2$

E $a > 0, b > 0, \max(a^2, b^2) < c < a^2 + b^2$

Worked solution

Answer A

Completing the square gives centre $(-a, -b)$ and radius squared $R^2 = a^2 + b^2 - c$. Strict containment in the first quadrant requires $-a > R$ and $-b > R$, so $a, b < 0$, $c > a^2, b^2$, and $c < a^2 + b^2$ for a nondegenerate circle. This is option A.

Question 25

A geometric progression has first term $a > 0$ and real common ratio r . Both infinite series converge, and $\sum_{n=1}^{\infty} ar^{n-1} = 2$, while $\sum_{n=1}^{\infty} (ar^{n-1})^2 = 12$. What is $\sum_{n=1}^{\infty} |ar^{n-1}|$?

A 4

B 6

C 2

D 12

E 8

Worked solution

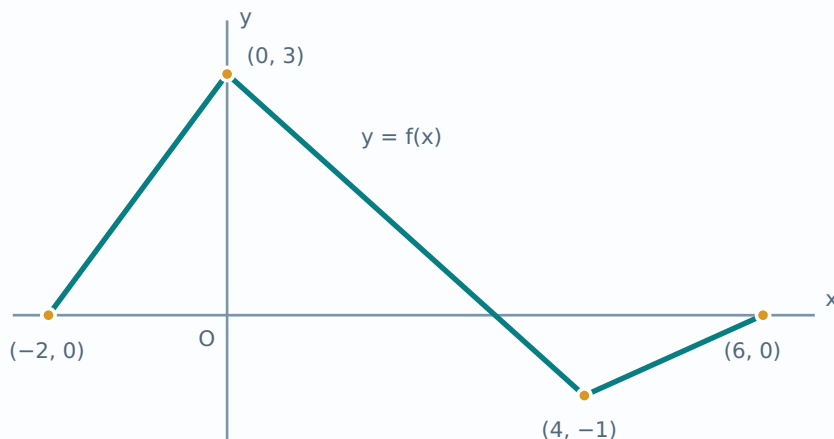
Answer B

Convergence gives $|r| < 1$, $a/(1-r) = 2$, and $a^2/(1-r^2) = 12$. Substitute $a = 2(1-r)$: $4(1-r)/(1+r) = 12$, so $r = -1/2$ and $a = 3$. The absolute values form a geometric series with first term 3, ratio $1/2$, and sum 6.

Trap: assuming r is positive; the signed sum forces a negative ratio.

Question 26

The function f is defined for $-2 \leq x \leq 6$ by the graph shown. For which values of x is $2f(3-x) + 1 < 0$?



The graph consists of straight line segments joining the labelled points.

A $-3 < x < -1$

B $-2 < x < -\frac{1}{2}$

C $-1 < x < \frac{1}{2}$

D $\frac{1}{2} < x < 2$

E $\frac{7}{2} < x < 5$

Worked solution

Answer B

First rearrange the inequality to $f(3-x) < -\frac{1}{2}$. Write $t = 3-x$ for the input to f .

On the descending segment from $(0, 3)$ to $(4, -1)$, the graph has equation $f(t) = 3 - t$, which reaches $-\frac{1}{2}$ at $t = \frac{7}{2}$. On the rising segment from $(4, -1)$ to $(6, 0)$, its equation is $f(t) = \frac{t-6}{2}$, which reaches $-\frac{1}{2}$ at $t = 5$. The graph is below this level precisely when

$$\frac{7}{2} < t < 5.$$

Substitute $t = 3-x$: $\frac{7}{2} < 3-x < 5$. Subtracting 3 and reversing the inequalities on multiplying by -1 gives $-2 < x < -\frac{1}{2}$. These inputs are within the domain of the transformed function.

Question 27

A geometric progression has positive terms. The sum of its first four terms is 30, and the sum of its first eight terms is 510. What is its sixth term?

A 32

B 48

C 60

D 64

E 128

Worked solution

Answer D

Let the first term be $a > 0$ and the common ratio be $r > 0$. Terms five to eight are r^4 times terms one to four. Their sum is $510 - 30 = 480$, so

$$30r^4 = 480, \quad r^4 = 16.$$

All terms are positive, so $r = 2$. The first four terms sum to $a(1 + 2 + 4 + 8) = 15a = 30$, giving $a = 2$. Therefore the sixth term is $ar^5 = 2 \cdot 2^5 = \boxed{64}$.

MODULE 03

Biology

QUESTIONS & WORKED SOLUTIONS

Maths 2 + Biology pathway

27

questions

40

minutes

1

mark per correct answer

NO CALCULATOR

Using this module

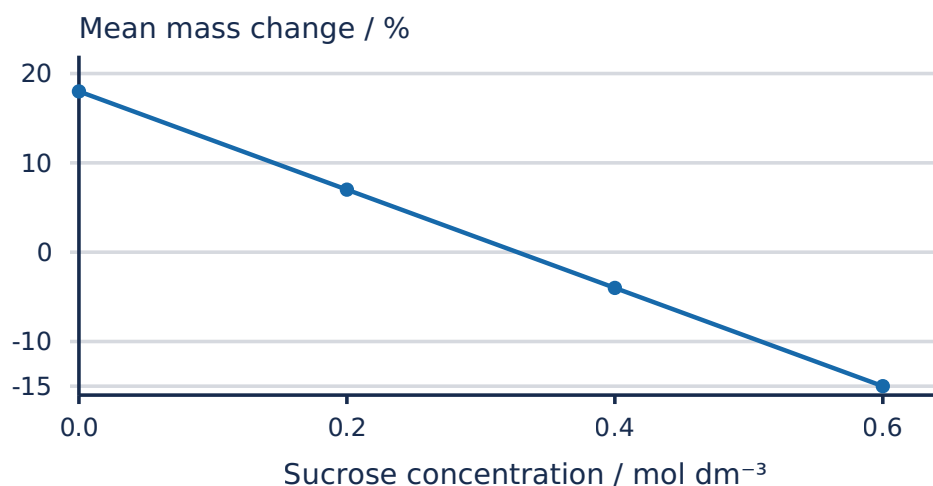
Choose one answer per question. There is no negative marking.
Each worked solution appears immediately below its question.
Attempt the question first; cover the solution if working to time.
Use separate paper for your working.

Question 1 begins on the next page.

Question 01

Potato cylinders were placed in sucrose solutions for 2 hours. The mean percentage change in mass was measured.

Sucrose / mol dm ⁻³	0.0	0.2	0.4	0.6
Change in mass / %	+18	+7	-4	-15



Assuming the relationship is linear between the two concentrations that bracket zero change, which external sucrose concentration would give zero mean percentage change in mass?

A 0.20 mol dm⁻³

B 0.27 mol dm⁻³

C 0.33 mol dm⁻³

D 0.45 mol dm⁻³

E 0.60 mol dm⁻³

Worked solution

Answer C

Zero change is between 0.2 and 0.4 mol dm⁻³. The measured change falls by 11 percentage points over this interval; reaching zero requires a fall of 7.

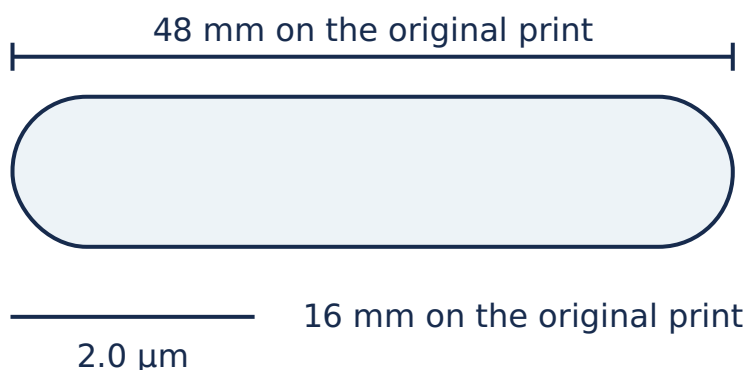
Concentration = $0.2 + (7/11) \times 0.2 = 0.327... \text{ mol dm}^{-3}$, giving **C, 0.33 mol dm⁻³**.

This estimates the external concentration giving no net mass change under the experiment's conditions.

Useful approach: Interpolate between the two measurements on opposite sides of zero.

Question 02

On an original printed micrograph, a scale bar labelled 2.0 μm measures 16 mm and a bacterium measures 48 mm. These measurements are shown schematically below; do not measure this drawing.



Which row gives the magnification of the original print, the actual bacterium length and a structure present in a typical bacterium?

A 8.0×10^2 ; 6.0 μm ; a cell wall

B 8.0×10^3 ; 6.0 μm ; a nucleus

C 8.0×10^4 ; 6.0 μm ; a nucleus

D 8.0×10^3 ; 0.6 μm ; a cell wall

E 8.0×10^3 ; 6.0 μm ; a cell wall

Worked solution

Answer E

The scale-bar length is 16 mm = 16 000 μm , so magnification = $16\,000 / 2.0 = 8.0 \times 10^3$.

The bacterium is $48/16 = 3$ times the scale-bar length, so its actual length is $3 \times 2.0 = 6.0 \mu\text{m}$. A typical bacterium has a cell wall and lacks a membrane-bound nucleus. Therefore **E** is correct.

Useful approach: Magnification uses the stated original-print measurement; actual length can be found from the scale-bar ratio.

Question 03

Nitrate uptake by root tissue was measured for 30 minutes. The external nitrate concentration was kept constant and greater than the concentration inside the cells.

Condition	Uptake / $\mu\text{mol g}^{-1} \text{h}^{-1}$
Normal oxygen, no inhibitor	18
Low oxygen, no inhibitor	8
Normal oxygen, respiratory inhibitor	5
Low oxygen, respiratory inhibitor	4

Consider the following statements.

1. Uptake decreases when oxygen availability is reduced.
2. The results support an energy-dependent component of nitrate uptake.
3. Reducing oxygen has the same absolute effect on uptake with and without the inhibitor.

Which statements are correct?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

Worked solution

Answer E

Uptake falls when oxygen is reduced: 18 to 8 without inhibitor, and 5 to 4 with inhibitor. Statement 1 is true. Reduced uptake with low oxygen or a respiratory inhibitor supports an energy-dependent component, so statement 2 is true.

The absolute decreases are 10 and 1 $\mu\text{mol g}^{-1} \text{h}^{-1}$, which are different. Statement 3 is false. Hence **E: 1 and 2 only**.

Useful approach: Compare the change within each inhibitor condition, rather than comparing only final values.

Question 04

A diploid organism has $2n = 6$. Count a chromosome by its centromere and count one chromatid as one DNA molecule. Immediately after DNA replication, a cell enters meiosis I. How many chromosomes and DNA molecules are present in each daughter cell immediately after meiosis I?

A 3 chromosomes and 3 DNA molecules

B 3 chromosomes and 6 DNA molecules

C 6 chromosomes and 6 DNA molecules

D 6 chromosomes and 3 DNA molecules

E 3 chromosomes and 12 DNA molecules

Worked solution

Answer B

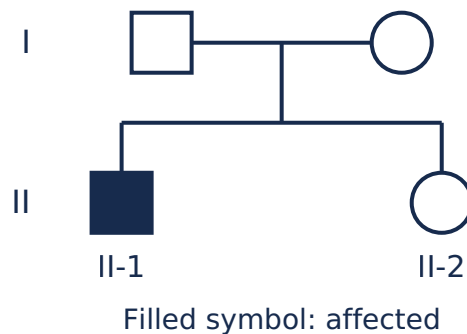
Before replication the cell has six chromosomes. Replication makes two DNA molecules (two chromatids) per chromosome, so there are 12 DNA molecules but still six chromosomes.

Meiosis I separates homologous chromosomes, not sister chromatids. Each daughter therefore receives three chromosomes, each still consisting of two chromatids: $3 \times 2 = 6$ DNA molecules. The answer is **B**.

Useful approach: Ask what is separating at each division. Meiosis I separates homologues; meiosis II separates sister chromatids.

Question 05

A rare condition is caused by an autosomal recessive allele. Two unaffected parents have an affected son and an unaffected daughter, II-2. II-2 has a child with an unrelated unaffected man. For this man, the probability of being a carrier is $1/10$. What is the probability that their child is affected?



Assume II-2 is unaffected as stated and that the father of her child is not related to her family.

A $1/10$

B $1/20$

C $1/40$

D $1/60$

E $1/120$

Worked solution

Answer D

Because the unaffected parents have an affected child, both are carriers. Their unaffected daughter has a $2/3$ probability of being a carrier: the three unaffected genotypes from a carrier $\times$ carrier cross are AA, Aa and Aa.

For an affected child, II-2 must be a carrier, the man must be a carrier, and two carrier alleles must meet:

$$(2/3) \times (1/10) \times (1/4) = 1/60.$$

Therefore **D** is correct.

Useful approach: Condition the daughter's genotype on being unaffected before multiplying by the unrelated partner's carrier risk.

Question 06

In a plant, purple flowers (P) are dominant to white flowers (p). A purple plant A is crossed with a homozygous white plant, and a large offspring sample contains equal numbers of purple and white plants. Plant A is then crossed with a known heterozygous purple plant B.

What is the probability that exactly two of the next three offspring are white?

A 9/64

B 3/16

C 1/16

D 1/4

E 27/64

Worked solution

Answer A

The first cross is $A \times pp$ and gives equal purple and white offspring, so A must be Pp. Plant B is also Pp. Each offspring is white with probability $1/4$.

Exactly two whites in three births can occur in three orders. Hence:

$$3 \times (1/4)^2 \times (3/4) = 9/64.$$

The answer is **A**; the factor of 3 is missed if one specific order is counted.

Useful approach: Count the three possible positions of the non-white offspring as well as the probability of one order.

Question 07

The coding DNA strand of a gene is shown below. The first triplet is the only part that changes in either variant.

normal: 5'-GAA CCT TGC-3'

variant 1: 5'-GAG CCT TGC-3'

variant 2: 5'-GAT CCT TGC-3'

Use the information that GAA and GAG code for glutamic acid, whereas GAT codes for aspartic acid. Which statement is correct?

- A** Both variants change the first amino acid, but neither changes the reading frame.
- B** Variant 1 leaves the first amino acid unchanged; variant 2 changes it.
- C** Variant 1 changes the first amino acid; variant 2 leaves it unchanged.
- D** Variant 1 shifts the reading frame; variant 2 changes only one amino acid.
- E** Both variants leave the entire amino-acid sequence unchanged.

Worked solution

Answer B

Variant 1 changes GAA to GAG. The supplied code says both specify glutamic acid, so it is a base substitution with no change to that amino acid (a silent substitution).

Variant 2 changes GAA to GAT, which changes glutamic acid to aspartic acid. It is therefore a missense substitution. Neither change adds or removes a base, so neither is a frameshift. **B** is correct.

Useful approach: Check whether the number of bases changes before considering a frameshift; then translate only the codon that changed.

Question 08

A gene and a plasmid are cut with a restriction endonuclease that leaves sticky ends. The diagram shows two ends that can be joined; 5' and 3' indicate strand direction.



Only the DNA near each cut end is shown.

Which sequence of events joins these ends?

- A** The overhangs pair by hydrogen bonds; a restriction endonuclease then joins the backbones.
- B** The overhangs pair by covalent bonds; DNA ligase then joins the backbones.
- C** DNA ligase first hydrogen-bonds the backbones; complementary bases then form covalent bonds.
- D** The overhangs pair by hydrogen bonds; DNA ligase then forms covalent bonds in the backbones.
- E** DNA polymerase pairs the overhangs; a restriction endonuclease then removes the paired bases.

Worked solution

Answer D

The complementary A-T base pairs form hydrogen bonds. The two strands run in opposite directions, as shown by their 5' and 3' ends.

DNA ligase seals the remaining breaks by forming covalent bonds in the sugar-phosphate backbones. The correct sequence is **D**. A restriction endonuclease cuts DNA; it does not seal the backbone.

Useful approach: Distinguish bonds between paired bases from bonds within a DNA strand.

Question 09

Matched samples of a bacterial population are tested before and after antibiotic exposure. Colony counts measure viable bacteria of each type.

	Susceptible	Resistant
Before exposure	9400	600
After exposure	50	550

Which row gives the resistant fraction among the surviving bacteria and the process that best explains its change?

- A** 5.5%; selection favouring resistant bacteria already present.
- B** About 92%; preferential survival of susceptible bacteria.
- C** About 92%; susceptible bacteria acquiring resistance because they need it.
- D** 6%; selection favouring resistant bacteria already present.
- E** About 92%; selection favouring resistant bacteria already present.

Worked solution

Answer E

There are $50 + 550 = 600$ surviving bacteria, of which 550 are resistant. The resistant fraction is $550/600 = 11/12 \approx 92\%$, compared with $600/10\,000 = 6\%$ before exposure.

Resistant variants were present initially and survived much more successfully. Selection therefore raises their frequency even though their absolute number decreases. The answer is **E**.

Useful approach: Use the total surviving population as the denominator, and distinguish frequency from absolute count.

Question 10

Equal enzyme samples are pre-treated for 10 minutes and then assayed with the same substrate concentration and temperature. The rate is measured immediately after transfer to the assay pH.

Pre-treatment pH	Assay pH	Rate / arbitrary units
6	6	14
9	9	2
9	6	13

Which comparison and interpretation fit the results?

- A** The first rate is 7 times the second; most activity lost at pH 9 is recovered at pH 6.
- B** The first rate is 7 times the second; pH 9 irreversibly denatures most enzyme molecules.
- C** The first rate is 7 times the second; the higher rate is caused by a higher assay substrate concentration.
- D** The second rate is 7 times the first; returning to pH 6 restores enzyme activity.
- E** The third rate is half the first; pre-treatment at pH 9 prevents recovery of activity.

Worked solution

Answer A

The rate ratio for the first two samples is $14/2 = 7$. After pH 9 pre-treatment, transfer to pH 6 gives a rate of 13, close to the control rate of 14.

Most activity therefore returns when pH is restored. The low rate at pH 9 does not imply irreversible denaturation of most enzyme molecules. Substrate concentration was held constant. Hence **A**.

Useful approach: Use the recovery experiment to distinguish reversible loss of activity from irreversible damage.

Question 11

Three membrane bags contain equal amounts of starch and protein, but different enzyme preparations. Each bag is placed in water at the same temperature and suitable pH.

The membrane allows maltose and amino acids through, but not starch, protein or enzymes. After 20 minutes, the surrounding water gives these results:

Preparation	Maltose detected	Amino acids detected
P	Yes	No
Q	No	Yes
R	Yes	Yes

Assume the preparations contain only amylase, protease, or both. Which row identifies P and Q and explains how products reach the surrounding water?

- A** P: protease; Q: amylase; products diffuse through the membrane.
- B** P: amylase; Q: protease; enzymes diffuse out and digest food outside the bags.
- C** P: amylase; Q: protease; soluble products diffuse through the membrane.
- D** P: amylase; Q: both enzymes; products pass through by osmosis.
- E** P: both enzymes; Q: protease; the membrane actively transports digestion products.

Worked solution

Answer C

Amylase digests starch to sugars including maltose, so P contains amylase. Protease digests protein to smaller products including amino acids, so Q contains protease. R contains both.

The permitted small soluble products diffuse from the bags into the surrounding water. Enzymes and undigested food molecules cannot cross the stated membrane. Therefore **C** is correct.

Useful approach: Separate digestion of large molecules from movement of the resulting small molecules.

Question 12

An athlete exercises from 5 to 20 minutes. Blood lactate concentration is measured alongside oxygen uptake.

Time / min	0	5	10	15	20	25	30
O ₂ uptake / L min ⁻¹	0.3	1.8	2.4	2.6	2.6	1.2	0.5
Lactate / mmol dm ⁻³	1.0	1.2	2.0	4.8	6.0	5.0	2.1

During which five-minute interval is the net increase in lactate concentration greatest, and what does the full pattern indicate?

- A** 5–10 minutes; anaerobic respiration stops as oxygen uptake rises.
- B** 15–20 minutes; the highest concentration marks the fastest lactate accumulation.
- C** 10–15 minutes; all respiration during exercise is anaerobic.
- D** 10–15 minutes; anaerobic respiration contributes while aerobic respiration continues.
- E** 20–25 minutes; lactate continues accumulating after exercise stops.

Worked solution

Answer D

The five-minute lactate changes are +0.2, +0.8, +2.8, +1.2, –1.0 and –2.9 mmol dm⁻³. The largest increase is therefore during **10–15 minutes**.

Oxygen uptake is elevated while lactate accumulates, so aerobic respiration continues and anaerobic respiration also contributes. A high final concentration need not mean the fastest rate of accumulation. The answer is **D**.

Useful approach: For equal time intervals, compare changes in concentration rather than final concentrations.

Question 13

A subject's cardiac output was measured at different heart rates.

Heart rate / beats min ⁻¹	60	80	100	120
Cardiac output / dm ³ min ⁻¹	4.8	6.0	7.0	6.6

What are the stroke volumes at heart rates of 80 and 120 beats min⁻¹, respectively?

A 7.5 cm³ and 5.5 cm³

B 75 cm³ and 55 cm³

C 75 cm³ and 82.5 cm³

D 80 cm³ and 55 cm³

E 750 cm³ and 550 cm³

Worked solution

Answer B

Stroke volume = cardiac output / heart rate.

At 80 beats min⁻¹: $6.0/80 = 0.075 \text{ dm}^3 = 75 \text{ cm}^3$.

At 120 beats min⁻¹: $6.6/120 = 0.055 \text{ dm}^3 = 55 \text{ cm}^3$. Therefore **B**. Higher heart rate does not necessarily mean greater stroke volume.

Useful approach: Divide output by the matching heart rate, then convert dm³ to cm³.

Question 14

For this question, a gas diffuses from higher to lower partial pressure. A larger exchange surface increases its rate of diffusion when other factors are unchanged.

The following partial pressures are measured in a healthy person.

Location	pO ₂ / kPa	pCO ₂ / kPa
Alveolar air	13.3	5.3
Blood arriving at alveolar capillaries	5.3	6.1

Consider the following statements.

1. Oxygen diffuses from alveoli to blood.
2. Carbon dioxide diffuses from blood to alveoli.
3. Reducing alveolar surface area would slow both transfers, all else equal.

Which statements are correct?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

Worked solution

Answer H

Oxygen moves down its partial-pressure gradient from 13.3 kPa in alveolar air to 5.3 kPa in arriving blood. Carbon dioxide moves from 6.1 kPa in blood to 5.3 kPa in alveolar air. A smaller exchange surface reduces diffusion rate if other factors are unchanged. Therefore all three statements are correct: **H**.

Useful approach: Compare oxygen and carbon dioxide separately; their gradients point in opposite directions.

Question 15

Compare the same healthy person after several hours without water with that person after drinking a large volume of water. Which row describes the dehydrated state **relative to the state after drinking**?

Row	ADH level	Collecting-duct water permeability	Urine volume	Urine concentration
A	Higher	Greater	Larger	Lower
B	Higher	Lower	Smaller	Higher
C	Higher	Greater	Smaller	Higher
D	Lower	Lower	Larger	Lower
E	Lower	Greater	Smaller	Higher

A Row A

B Row B

C Row C

D Row D

E Row E

Worked solution

Answer C

Dehydration lowers blood water potential, stimulating ADH release. Higher ADH increases the collecting ducts' permeability to water, so more water is reabsorbed.

Urine volume is consequently smaller and the urine is more concentrated than after drinking. All four changes agree only in **row C**.

Useful approach: Follow the sequence: blood water potential → ADH → water reabsorption → urine.

Question 16

Person P eats a carbohydrate-rich meal. Their blood glucose rises from 4.5 to 9.0 mmol dm⁻³ and remains high for several hours. Plasma insulin remains low. An injection of insulin is followed by a fall in blood glucose. Assume the target cells respond normally to insulin.

Which inference is best supported?

- A** Target cells are insensitive to insulin; extra insulin lowers glucose mainly by stimulating glycogen breakdown.
- B** Target cells are insensitive to insulin; the low measured insulin concentration excludes a secretion problem.
- C** Insulin secretion is excessive; the injection lowers glucose by further increasing secretion.
- D** Insulin secretion is insufficient; the injection lowers glucose mainly by increasing glycogen breakdown.
- E** Insulin secretion is insufficient; the injection promotes glucose uptake and storage as glycogen.

Worked solution

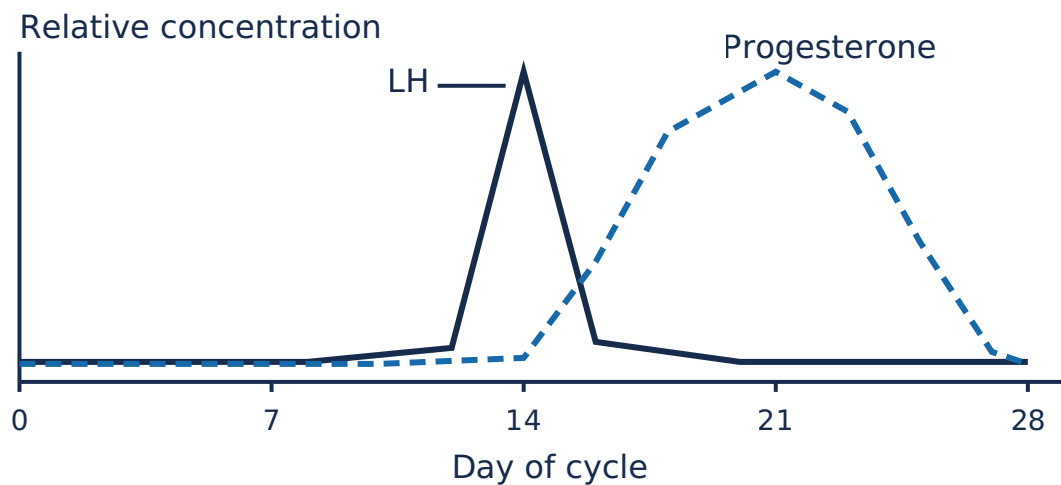
Answer E

Glucose remains high while endogenous insulin is low, but an insulin injection lowers it. With normal target-cell response, the most direct explanation is insufficient insulin secretion. Insulin promotes uptake of glucose by cells and conversion of glucose to glycogen, lowering blood concentration. **E** is correct.

Useful approach: Use both the measured hormone level and the response to the injection.

Question 17

The graph shows a typical non-pregnant menstrual cycle. Each hormone is plotted relative to its own maximum concentration.



Which statement correctly connects the hormone changes around days 14–21 with events in the ovary and uterus?

- A** The progesterone rise triggers ovulation; LH then maintains the uterine lining.
- B** The LH surge causes the uterine lining to break down; progesterone stimulates a new follicle.
- C** The LH surge releases the ovum; progesterone is then secreted by the pituitary.
- D** The LH surge triggers ovulation; progesterone from the corpus luteum helps maintain the uterine lining.
- E** The progesterone rise stimulates further LH release, allowing repeated ovulation during the same cycle.

Worked solution

Answer D

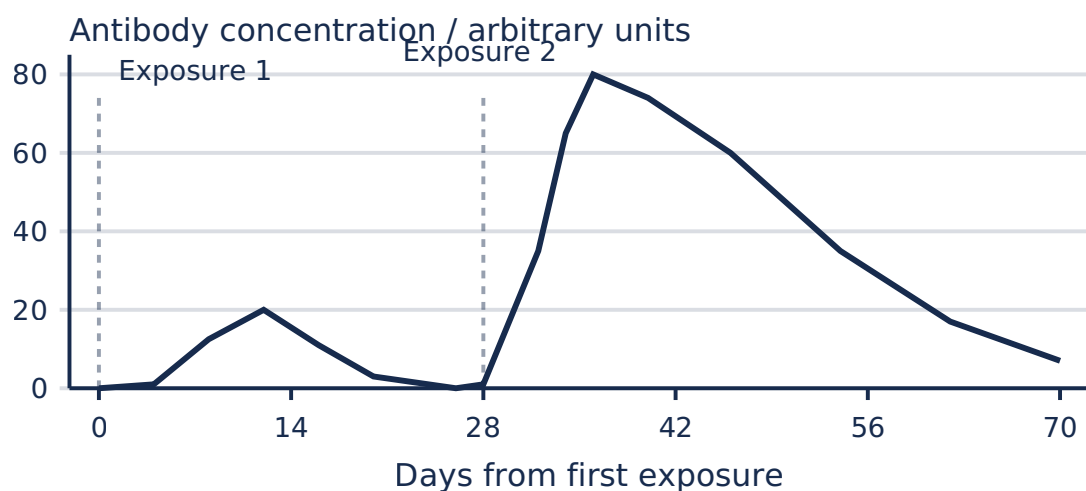
The sharp LH surge around day 14 triggers ovulation. The ruptured follicle develops into the corpus luteum, which secretes progesterone.

Progesterone helps maintain the uterine lining and contributes to negative feedback that suppresses further gonadotrophin release. Thus **D** is correct.

Useful approach: Link each hormone to its source and target rather than treating the two curves as interchangeable.

Question 18

A person receives the same dose of antigen X on days 0 and 28. The graph shows the concentration of antibodies against X.



What is the ratio of the second peak to the first, and which mechanism best explains the stronger second response?

- A** 4 : 1; antigen-specific memory lymphocytes respond on re-exposure.
- B** 2 : 1; antibodies remaining from the first response reproduce when they bind antigen.
- C** 2 : 1; antigen-specific memory lymphocytes respond on re-exposure.
- D** 4 : 1; pre-existing antibodies change their specificity after the second exposure.
- E** 4 : 1; non-specific phagocytes produce the increased antibody concentration.

Worked solution

Answer A

The peaks read from the graph are 20 and 80 arbitrary units, giving $80 : 20 = 4 : 1$.

The first exposure forms antigen-specific memory lymphocytes. Re-exposure activates a faster, larger response producing the relevant antibodies. Antibodies do not reproduce or change specificity, and phagocytes are not the cells that produce antibodies. Therefore **A**.

Useful approach: Read both peak heights from the same scale before choosing the biological explanation.

Question 19

A plant's net oxygen production was measured under four combinations of light intensity and carbon-dioxide concentration. Temperature and other conditions were kept constant.

	Low CO ₂	High CO ₂
Low light	4	5
High light	5	12

Which statement best interprets the interaction?

- A** At low light, carbon dioxide is the principal limiting factor because increasing it raises production from 4 to 5.
- B** At low carbon dioxide, raising light doubles net oxygen production.
- C** At high carbon dioxide, light is no longer limiting because increasing it changes production from 5 to 12.
- D** The effects of extra light and extra carbon dioxide are independent, because either alone changes 4 to 5.
- E** At high light, raising carbon dioxide gives a much larger increase; light limits the response to carbon dioxide at low light.

Worked solution

Answer E

At low light, raising CO₂ changes oxygen production only from 4 to 5, so light is the main limiting factor. At high light, raising CO₂ changes the rate from 5 to 12, showing that CO₂ becomes limiting once more light is available. This is **E**.

Useful approach: Compare the effect of one factor at each fixed level of the other factor.

Question 20

In a potometer, water uptake moves an air bubble through a uniform capillary tube. If the bubble moves distance h in time t , the volume uptake rate is $\pi r^2 h/t$, where r is the tube radius.

Potometer A



$$h = 8 \text{ mm}$$

$$r = 0.5 \text{ mm}$$

Potometer B



$$h = 4 \text{ mm}$$

$$r = 1.0 \text{ mm}$$

Solid bubble: start; dashed bubble: after 4 minutes.

What is the ratio of the water uptake rate in B to that in A?

A 1/4

B 1/2

C 1

D 2

E 4

Worked solution

Answer D

The measurement time is the same, so π and t cancel in the ratio:

$$B/A = (1.0^2 \times 4)/(0.5^2 \times 8) = 4/2 = \mathbf{2}, \text{ giving } \mathbf{D}.$$

Doubling the radius quadruples the cross-sectional area; halving the bubble displacement then halves this factor.

Useful approach: Bubble distance alone is not a volume: multiply by the capillary cross-sectional area.

Question 21

A complete ring of bark is removed from a tree trunk, leaving the inner wood and xylem vessels intact. Several weeks later, sugars have accumulated above the ring and the roots are losing mass.

Which explanation best accounts for both observations?

- A** Removing the bark interrupts xylem flow to the leaves; water accumulates above the ring as sugar.
- B** Removing the bark interrupts phloem transport toward the roots; xylem water transport can continue.
- C** Removing the bark interrupts phloem transport toward the leaves; the roots export their sugar supply upward.
- D** Removing the bark leaves phloem intact but prevents sugar from moving between xylem vessels.
- E** Removing the bark stops water uptake first; the intact phloem carries the roots' sugars upward.

Worked solution

Answer B

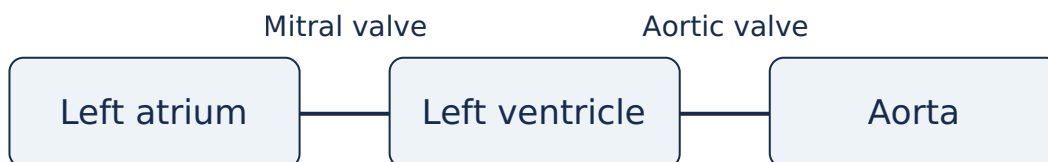
Phloem lies in the bark and transports assimilates from sources such as leaves to sinks such as roots. Removing a ring interrupts this downward translocation, causing sugar accumulation above the ring and reduced supply to roots.

The xylem is in the inner wood and has been left intact, so upward water transport can continue. This supports **B**.

Useful approach: Locate the tissue before naming the function: bark removal targets phloem, whereas the inner wood contains xylem.

Question 22

The diagram shows part of the left side of the heart. A valve opens when pressure on its upstream side exceeds pressure on its downstream side; otherwise it is closed. Ignore any delay in valve movement.



Normal permitted flow is from left to right.

Instant	Left atrium / kPa	Left ventricle / kPa	Aorta / kPa
I	1.0	0.6	10.0
II	1.0	5.0	10.0
III	1.0	13.0	12.0
IV	1.0	8.0	10.0

At which two instants are **both** valves closed?

A II and IV

B II and III

C I and II

D III and IV

E I and IV

Worked solution

Answer A

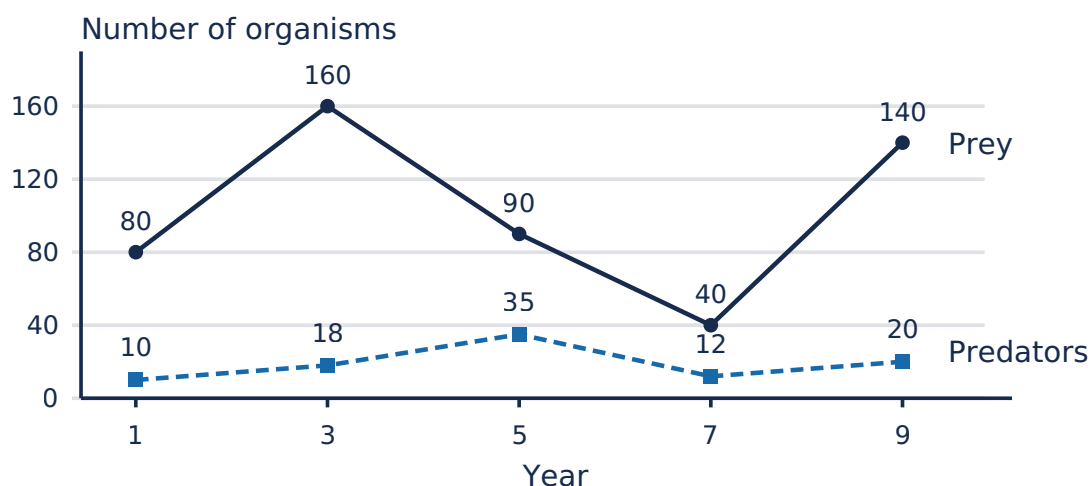
The mitral valve is closed when ventricular pressure exceeds atrial pressure. The aortic valve is closed when aortic pressure exceeds ventricular pressure.

Both conditions hold at II ($1 < 5 < 10$) and IV ($1 < 8 < 10$). At I the mitral valve is open; at III the aortic valve is open. Therefore **A: II and IV**.

Useful approach: Compare pressure across each valve separately; do not decide from ventricular pressure alone.

Question 23

Predator and prey populations are counted every two years in a closed ecosystem. Numbers beside the plotted points are the exact counts.



Consider these statements about the recorded counts.

1. The predator maximum occurs two years after the prey maximum.
2. The percentage fall in prey from year 3 to year 5 is greater than from year 5 to year 7.
3. The predator-to-prey ratio is greatest at year 5.

Which statements are correct?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

Worked solution

Answer F

Prey peaks at year 3 and predators at year 5, so statement 1 is true.

The two prey decreases are $70/160 = 7/16$ (less than one-half) and $50/90 = 5/9$ (more than one-half). Statement 2 is false.

Predator/prey ratios are $1/8$, $9/80$, $7/18$, $3/10$ and $1/7$. The largest is $7/18$ at year 5, so statement 3 is true. Hence **F: 1 and 3 only**.

Useful approach: A larger absolute decrease need not be a larger percentage decrease; use the starting population.

Question 24

Ten quadrats, each 0.25 m^2 , are placed at random in a meadow. The total number of plants of species A counted across all quadrats is 80. Which pair gives the estimated density and the best reason for placing the quadrats randomly?

- A** 8 plants m^{-2} ; random positions reduce investigator selection bias.
- B** 32 plants m^{-2} ; random positions reduce investigator selection bias.
- C** 32 plants m^{-2} ; random positions ensure every patch of meadow is sampled equally.
- D** 320 plants m^{-2} ; random positions reduce investigator selection bias.
- E** 20 plants m^{-2} ; random positions are chosen by starting at the densest patch.

Worked solution

Answer B

Total sampled area = $10 \times 0.25 = 2.5 \text{ m}^2$. Estimated density = $80 \div 2.5 = 32 \text{ plants m}^{-2}$. Random placement reduces the risk that the investigator chooses unusually dense or sparse patches, so **B** is correct.

Useful approach: Divide total plants by total sampled area; random placement reduces bias but cannot guarantee equal coverage.

Question 25

The table gives the energy incorporated into **new biomass each year** at the trophic levels shown for a grassland.

Trophic level	Producer	Primary consumer	Secondary consumer	Tertiary consumer
Annual production / $\text{kJ m}^{-2} \text{ year}^{-1}$	9000	1500	300	?

The energy-transfer efficiency from secondary-consumer production to tertiary-consumer production is 20%. Which row gives the missing value and a process contributing to reduced energy transfer between trophic levels?

- A** $6 \text{ kJ m}^{-2} \text{ year}^{-1}$; organic molecules are used in respiration.
- B** $60 \text{ kJ m}^{-2} \text{ year}^{-1}$; all material eaten is converted into new consumer tissue.
- C** $60 \text{ kJ m}^{-2} \text{ year}^{-1}$; some energy from respiration is transferred to the surroundings as heat.
- D** $150 \text{ kJ m}^{-2} \text{ year}^{-1}$; organic molecules are used in respiration.
- E** $1500 \text{ kJ m}^{-2} \text{ year}^{-1}$; some ingested material is egested without assimilation.

Worked solution

Answer C

Tertiary-consumer production is $0.20 \times 300 = 60 \text{ kJ m}^{-2} \text{ year}^{-1}$.

Some consumed material is not assimilated. Of the energy that is assimilated, some is released by respiration and ultimately transferred to the surroundings as heat instead of being retained in new biomass. Thus **C** gives both the value and a valid loss process.

Useful approach: Transfer efficiency compares production over the same period, rather than standing biomass measured at one instant.

Question 26

A mutation prevents a patient's liver cells from making protein P. Treatment inserts a functional gene into the patient's own stem cells and uses their descendants to replace some liver cells. A small amount of P released into the blood is sufficient. Gametes are not treated.

Which statement describes successful tissue replacement and the inheritance of this correction?

- A** The cells need to contain the gene, but need not express it; the correction is not normally inherited.
- B** The functional gene must be inserted into gametes before treated liver cells can express it; the correction is then inherited.
- C** The cells must differentiate into liver cells and express the gene; the correction is normally inherited through the patient's gametes.
- D** The cells must remain undifferentiated to replace liver cells; the correction is not normally inherited.
- E** The cells must differentiate into liver cells and express the gene; the correction is not normally inherited.

Worked solution

Answer E

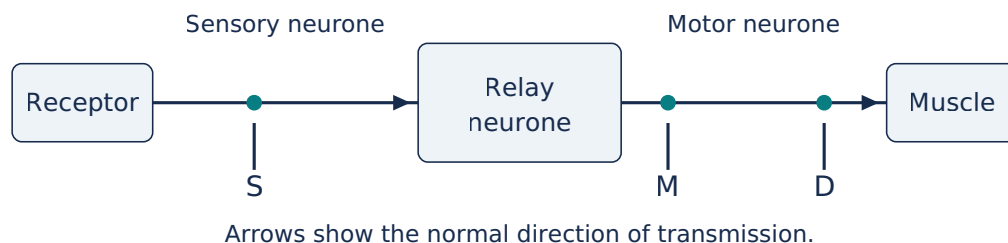
The stated aim is to replace liver cells, so the treated stem cells must produce the required differentiated descendants. Those cells must express the inserted gene to make functional protein P; carrying DNA alone is insufficient.

The correction is to somatic cells. Because the gametes are not treated, it is not normally passed to the patient's children. The answer is **E**.

Useful approach: Distinguish possession of a gene, expression of that gene and transmission through gametes.

Question 27

Electrical stimulation at S, M or D directly starts an impulse in the neurone at that point. In an intact preparation, stimulation at any of the three sites makes the muscle contract.



A single complete block to impulse transmission develops somewhere along the illustrated pathway. Stimulation does not spread to neighbouring cells. The following results are obtained:

Site stimulated	Muscle response
S	No contraction
M	No contraction
D	Contraction

Where is the block?

- A** In the motor neurone between M and D.
- B** At the connection between the relay neurone and the motor neurone.
- C** In the sensory neurone between S and the relay neurone.
- D** Between D and the muscle.
- E** Within the muscle's response to an arriving nerve impulse.

Worked solution

Answer A

Stimulation at D produces contraction, so transmission from D to the muscle and the muscle response remain functional.

Stimulation at M fails, placing the block between M and D. This also explains why stimulation at the more upstream site S fails. A block before M would be bypassed by direct stimulation at M. Therefore **A** is correct.

Useful approach: Start with the most downstream successful stimulus, then compare it with the nearest unsuccessful one.